



Investigation of neurofilament light chain in serum from the PPMI cohorts as a progression marker for PD

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Summary (or Abstract)

The analyte investigated in this project was **neurofilament light chain (NfL)**, which has growing interest in the neurodegenerative field. The 68-70 [kDa](#) NfL may have a significant biomarker role in differentiating PD from other parkinsonian disorders, which has been described in cerebrospinal fluid (CSF)¹ There are emerging highly sensitive methods for the quantification of NfL also in peripheral blood with good known correlation between CSF and blood NfL (own data). **NfL protein was measured in serum samples** from the PPMI sample collection (cross sectional and longitudinal) on the Simoa singleplex NF-light assay.





Method

The NF-light® assay is a 2-step digital immunoassay designed to measure the quantity of total NF-light in serum, plasma, and CSF using the fully automated Simoa HD-1 Analyzer and Single Molecule Array (Simoa) technology.

In the first step, anti-NF-light antibody coated paramagnetic capture beads, sample, and biotinylated detector antibody are combined in duplicate. PPMI serum samples were diluted 1:4. NF-light molecules present in the sample are captured by the anti-NF-light antibody coated capture beads and labeled with biotinylated detector antibodies. After washing, a conjugate of streptavidin- β -galactosidase (SBG) is mixed with the capture beads. SBG binds to the biotinylated detector antibodies, resulting in enzyme labeling of captured NF-light. Following a second wash, the capture beads are resuspended in a resorufin β -D-galactopyranoside (RGP) substrate solution and transferred to the Simoa Disc. Individual capture beads are then sealed within microwells in the array. If NF-light has been captured and labeled, the β -galactosidase hydrolyzes the RGP substrate into a fluorescent product that provides the signal for measurement. A single labeled NF-light molecule results in sufficient fluorescent signal in 30 seconds to be detected and counted by the Simoa optical system. At low NF-light concentration, the percentage of bead-containing wells in the array that have a positive signal is proportional to the amount of NF-light present in the sample. At higher NF-light concentration, when most of the bead-containing wells have one or more labeled NF-light molecules, the total fluorescence signal is proportional to the amount of NF-light present in the sample. The concentration of NF-light in unknown samples is interpolated from a standard curve. Sample results are reported as dilution corrected concentration. Overall, 5 out of 7 standards and controls, must have a cumulative mean accuracy (% Bias) $\pm 25\%$ of the nominal concentration and standards, controls and samples must have precision (%CV) of $\leq 25\%$ for the duplicate values to be considered acceptable.

<https://www.quanterix.com/products-technology/simoa-assay-kits>

References

1. Hu X, Yang Y, Gong D. Cerebrospinal fluid levels of neurofilament light chain in multiple system atrophy relative to Parkinson's disease: a meta-analysis. *Neurol Sci.* Mar 2017;38(3):407-414.
2. Thelin E, Nimer F, Frostell A, Zetterberg H, Blennow K, Nystrom H, Svensson M, Bellander B, Piehl F, Nelson DW. A Serum Protein Biomarker Panel Improves Outcome Prediction in Human Traumatic Brain Injury. *Journal of Neurotrauma*, 2019 May 9





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